B1 partial fine-tuning

Phase 4 Court Judgment Prediction | source-audited documentation

Problem statement and data

Phase 4 Court Judgment Prediction predicts binary outcome 0/1 from legal case text. The preserved official inputs are Train/train.json, Train/dev.json and Test/test.json. Test labels are private. Original run records report 994 official dev labels and 1,517 test IDs; macro-F1 is the central metric.

Architecture and mechanics

The original six-encoder/six-decoder B1 backbone is partially unfrozen. The shared evidence head pools tokens and models ordered chunks with a bidirectional GRU. Data flow: Legal case -> B1 own tokenizer/backbone -> Last two encoder and decoder blocks, plus norms -> Token evidence pooling -> Ordered chunk BiGRU -> 2-class logits -> prediction.json.

Training design and checkpoint selection

Last two encoder and decoder blocks, plus norms. Executed v3 runner: code/v3_t4x2.ipynb, code/runner/phase4_cjpe/kaggle_runner.py with published_v3_schedule.json; the six jobs share EvidenceCaseModel in code/runner/phase4_cjpe/models.py.

Exact run configuration and resource tradeoff

Experiment family b1; adaptation method partial; original parent checkpoint SHA-256 d93a2f1796e90b9fdbe4a7b542f76eb6065dab619feed4a1ef5a9952e2376a1f. Configuration: 256 tokens/chunk, 8 chunks/case, 8192 source characters considered, maximum 12000 training cases, 256 holdout cases per class. Optimizer configuration: backbone learning rate 2e-05, head learning rate 0.0003, gradient accumulation 8, max epochs 4, planned training hours 2.5. Recorded 1233 optimizer updates and 9864 seen training examples; selected step 1100. Selected checkpoint SHA-256 4c4ca537ec9a02bac12b577cf73b5fb7db29fff9eb175b15103a30bd291f1977; parameters 125274250, trainable 34032778. Original run completed after nominal deadline: True.

Measured results

Original Kaggle run: complete; 1233 optimizer updates; selected step 1100. Official 994-case dev macro-F1 0.594530 at recorded threshold 0.730290; default 0.5 dev macro-F1 0.397437. Verified prediction ZIP SHA-256 d9acd7b64973e6e815f743003eca864ff7b569590bdf53fcef743f1cc996728a, exactly 1,517 IDs. User-provided Codabench screenshot: 0.5895 (finished submission).

Evaluation mechanics

Official labeled dev cases 994; class supports [497, 497]; confusion matrix rows true 0/1, columns predicted 0/1: [[399, 98], [290, 207]]. At selected threshold 0.730290, macro-F1 0.594530; at threshold 0.5, macro-F1 0.397437. The report keeps these thresholds separate because threshold choice changes the binary predictions. The submitted ZIP has 1,517 unique test IDs and binary labels; no test labels were available to this run.

Prediction contract and verification

The original prediction ZIP contains one root-level prediction.json with 1,517 unique official test IDs and integer labels restricted to 0 and 1. This package checks the ZIP CRC and SHA-256 against the archived run/verification record. Full model checkpoint tensors remain outside GitHub; their downloaded backup receipts and hashes live in the local Phase 4 study archives.

Reproducibility and limits

The local dev score uses a threshold recorded in the run; inspect its calibration/selection record before treating it as independent model-selection evidence. A prepared model notebook is source scaffolding, while v3_t4x2.ipynb and the Kaggle run evidence identify the executed v3 lineage. Reports and companion notebook were created after training; original scores come from copied artifacts. Model weights and course data are preserved locally, not uploaded to GitHub.